Sub-PRD 2 - Client Admin - Learner Management

Document Type: PRD — Detailed Feature Spec with Acceptance Criteria

Parent Document: Client Admin Master PRD

Depends On: Sub-PRD 1 — Client Admin Platform - Org Hierarchy & RBAC (Department, Team, Group entities must exist)

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Status: NOT STARTED

Version History:

Scope: Feature 2 (Learner Management) from Master PRD — full detail, all edge cases, complete acceptance criteria

> ⚠️ **V1 SCOPE LOCKED** — See Section 3 for explicit out-of-scope list.

> 📌 **Phone field: NOT required at learner creation (confirmed March 22, 2026). Full Name + Email are the only required fields.**

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1. Problem This PRD Solves

A coaching centre or corporate training client needs to populate their tenant with learners before any learning can happen.

This is the **most operationally frequent action** a Client Admin and Team Manager perform — every new batch intake, every new employee joining, every mid-year transfer triggers it.

The problem has three distinct failure modes seen in competing platforms:

1. **Manual add is tedious** — no validation, no duplicate detection, no confirmation of what was actually created
2. **CSV upload is fragile** — one bad row kills the entire import, no partial success, no downloadable error log
3. **Post-creation lifecycle is underpowered** — no clean deactivation, no learner profile editing, no visibility into a learner's current state

This PRD defines the **complete learner management lifecycle** — from creation to deactivation — with production-grade validation, a polished CSV pipeline, and full audit coverage.

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2. Locked Decisions (Pre-Writing)

Decision	Value
Required fields at creation	Full Name + Email only. Phone is OPTIONAL.
Optional fields at creation	Phone, Department, Team, Employee/Roll number, Notes
Credential flow	Client Admin manually sets password during creation
CSV password column	Present in template — if left blank, system auto-generates and exposes in result file
Duplicate detection key	Email + Phone — both must be unique within the tenant
Duplicate rule	If EITHER email OR phone already exists in the tenant → block with specific field error
Hard delete	V2 (GDPR scope) — V1 has deactivate/reactivate only
Who can deactivate/reactivate	Client Admin only
Who can add learners	Client Admin + Team Manager (TM adds to own dept only)
CSV limits	5 MB max file size; ≤100 rows → sync; >100 rows → async with email notification
CSV error handling	Row-by-row validation before commit; option to import valid rows only; error log downloadable as CSV

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3. Scope Boundaries

In Scope

- Manual learner add (single learner form)
- CSV bulk upload (template download, upload, validate, commit)
- Learner list view with search, filter, sort
- Learner profile view + edit
- Deactivate / reactivate learner
- Remove learner from Team
- Remove learner from Group
- Password reset by Client Admin
- Audit log events for all learner lifecycle actions
- All edge cases

Out of Scope (Deferred)

- Self-registration via invite link → V2
- SSO / SAML login → V2
- Hard delete (GDPR) → V2
- Learner impersonation → V2
- Learner-facing password reset (self-service) → V2
- Dual role (Client Admin + Learner same account) → V2
- Bulk move learners across departments → V2

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4. Learner Data Model

4.1 User Entity (shared with platform)

Field	Type	Required	Notes
`user_id`	UUID	System	Auto-generated
`full_name`	VARCHAR(150)	Yes	Trimmed; display name
`email`	VARCHAR(255)	Yes	Lowercase-normalised; unique within tenant

`phone`	VARCHAR(20)	No	E.164 format preferred; unique within tenant if provided
`password_hash`	VARCHAR	Yes	bcrypt; set by Client Admin at creation
`role`	ENUM	System	`B2B_LEARNER` at creation
`tenant_id`	UUID	System	Injected by middleware
`status`	ENUM	System	`ACTIVE` / `INACTIVE`
`created_by`	UUID	System	user_id of Client Admin or Team Manager
`created_at`	TIMESTAMP	System	UTC
`updated_at`	TIMESTAMP	System	UTC

4.2 LearnerProfile Entity (B2B-specific extension)

Field	Type	Required	Notes
`profile_id`	UUID	System	Auto-generated
`user_id`	UUID	System	FK → User
`tenant_id`	UUID	System	FK → Tenant
`employee_roll_number`	VARCHAR(100)	No	Optional identifier; unique within tenant if provided
`notes`	TEXT	No	Admin-only field; not visible to learner
`department_id`	UUID	No	FK → Department; NULL = Unassigned

`team_id`	UUID	No	FK → Team; NULL = no team

4.3 Uniqueness Constraints (DB-level)

```sql
-- Email unique within tenant
UNIQUE INDEX uq_user_email_tenant ON users (email, tenant_id);

-- Phone unique within tenant
UNIQUE INDEX uq_user_phone_tenant ON users (phone, tenant_id);

-- Roll number unique within tenant (when provided)
UNIQUE INDEX uq_roll_number_tenant ON learner_profiles (employee_roll_number, tenant_id)
  WHERE employee_roll_number IS NOT NULL;
```

Cross-tenant rule: Same email CAN exist in different tenants. A learner at two coaching centres has two separate accounts. This is by design — tenants are fully isolated.

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5. Manual Learner Add — Full Feature Spec

5.1 Entry Point

Client Admin: Learners (sidebar) → "+ Add Learner" button (top right)
Team Manager: Learners (sidebar) → "+ Add Learner" button → Department is pre-filled and locked to their own dept

5.2 Form Fields

Field	Type	Required	Validation
Full Name	Text	Yes	Min 2 chars; max 150 chars; trimmed
Email	Email	Yes	Valid email format; unique within tenant
Phone	Text	No	Min 7 digits; max 15 digits; numeric + `+` allowed; unique within tenant if provided
Password	Password	Yes	Min 8 chars; at least 1 uppercase, 1 number
Confirm Password	Password	Yes	Must match Password

Department	Dropdown	No	Active departments only; pre-filled + locked for Team Manager
Team	Dropdown	No	Active teams within selected Department; cleared if Department changes
Employee / Roll Number	Text	No	Max 100 chars; unique within tenant if provided
Notes	Textarea	No	Max 1000 chars; admin-only

5.3 Validation Behaviour

- All validation runs **inline on blur** (field loses focus) — not only on submit
- Duplicate email check fires on blur via API (`GET /api/learners/check-email?email=x`)
- Duplicate phone check fires on blur via API (`GET /api/learners/check-phone?phone=x`)
- Password strength indicator shown live as Client Admin types
- "Save" button is disabled until all required fields pass validation

5.4 On Successful Save

1. `User` record created with `status = ACTIVE` , `role = B2B_LEARNER`
2. `LearnerProfile` record created with department/team if provided
3. No email sent to learner (credential flow is admin-managed)
4. Client Admin sees success toast: *"Priya Sharma added successfully."*
5. Client Admin is redirected to the new learner's profile page
6. Audit log: `LEARNER_CREATED`

5.5 Acceptance Criteria

AC-2.1.1 — Successful manual add

```
Given I am logged in as Client Admin
And no user with email "priya@example.com" exists in my tenant
```

```
And no user with phone "9876543210" exists in my tenant
When I fill the Add Learner form:
    full_name     = "Priya Sharma"
    email         = "priya@example.com"
    phone         = "9876543210"
    password      = "Welcome@123"
    department_id = "NEET Batch 2026"
    team_id       = "Batch A"
And I click Save
Then a User record is created:
    full_name = "Priya Sharma"
    email     = "priya@example.com"
    phone     = "9876543210"
    status    = ACTIVE
    role      = B2B_LEARNER
    tenant_id = my tenant
And a LearnerProfile record is created:
    department_id = NEET Batch 2026
    team_id       = Batch A
And I am redirected to Priya's profile page
And audit entry: action = LEARNER_CREATED, actor = Client Admin
And Priya can log in with email "priya@example.com" + password "Welcome@123"
```

AC-2.1.2 — Duplicate email blocked

```
Given a user with email "priya@example.com" already exists in my tenant
When I enter email "priya@example.com" in the Add Learner form
And the field loses focus
Then an inline error appears:
    "This email is already registered in your organisation."
And the Save button remains disabled
```

AC-2.1.3 — Duplicate phone blocked

```
Given a user with phone "9876543210" already exists in my tenant
When I enter phone "9876543210" in the Add Learner form
And the field loses focus
Then an inline error appears:
    "This phone number is already registered in your organisation."
And the Save button remains disabled
```

AC-2.1.4 — Same email, different tenant is allowed

```
Given user "priya@example.com" exists in Tenant A (a different tenant)
And my tenant is Tenant B
When I add a learner with email "priya@example.com" in Tenant B
Then the learner is created successfully
And Tenant A's data is unaffected
```

AC-2.1.5 — Team Manager pre-fill

```
Given I am Team Manager for "NEET Batch 2026"
When I open the Add Learner form
Then the Department field is pre-filled with "NEET Batch 2026"
And the Department field is read-only (locked)
And I can only select Teams within "NEET Batch 2026"
And the created learner's department_id = NEET Batch 2026
```

AC-2.1.6 — Password rules enforced

```
Given I enter password "abc" in the Add Learner form
Then the password strength indicator shows "Weak"
```

```
And an inline error shows:
    "Password must be at least 8 characters with 1 uppercase letter and 1 number."
And the Save button remains disabled
```

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6. CSV Bulk Upload — Full Feature Spec

6.1 Entry Point

Learners (sidebar) → "+ Add Learner" → "Bulk Upload via CSV" tab

6.2 CSV Template

Client Admin downloads the template from the upload screen. Template filename: `keyskillset_learner_import_template.csv`

Template columns:

Column	Required	Notes
`full_name`	Yes	Max 150 chars
`email`	Yes	Valid email; unique within tenant
`phone`	No	Numeric + `+`; unique within tenant if provided
`password`	No	Min 8 chars, 1 uppercase, 1 number; if blank → system auto-generates
`department_name`	No	Must match an existing active Department name exactly (case-insensitive)
`team_name`	No	Must match an existing active Team name within the specified department
`employee_roll_number`	No	Unique within tenant if provided
`notes`	No	Max 1000 chars

Template rules:

- Row 1 = header (column names exactly as above — case-insensitive match)
- Row 2 onwards = data rows
- Empty rows are skipped silently
- BOM (byte order mark) is stripped automatically
- UTF-8 encoding required

6.3 Upload Flow

```
Step 1: Client Admin uploads CSV file
         → File size check: max 5 MB (error if exceeded)
         → File type check: .csv only (error if .xlsx or other)
         → Row count detected

Step 2: System validates every row before any commit
         → Structural validation (correct columns present)
         → Per-row field validation
         → Duplicate detection (email + phone) within the file
         → Duplicate detection against existing tenant learners
         → Department / Team name resolution

Step 3: Validation result screen shown:
         → Total rows: X
         → Valid rows: Y
         → Error rows: Z (with error details per row)
         → Two action buttons:
             "Import Valid Rows Only (Y rows)"
             "Cancel — Fix Errors First"
         → "Download Error Log" button (always visible if Z > 0)

Step 4: Client Admin clicks "Import Valid Rows Only" or cancels

Step 5 (≤100 valid rows): Synchronous import
         → All valid rows committed in a single transaction
         → Success screen: "Y learners added successfully."
         → Error log still downloadable

Step 5 (>100 valid rows): Asynchronous import
         → Job queued, job_id returned
         → Client Admin sees: "Your import is processing.
            We'll email you at [admin email] when complete."
         → Email notification sent on completion with:
             - Total imported: Y
             - Failed rows: Z (if any post-processing failures)
             - Downloadable result CSV attached
```

6.4 Per-Row Validation Rules

Check	Error Message
`full_name` missing	Row {N}: full_name is required
`email` missing	Row {N}: email is required
`email` invalid format	Row {N}: email "{value}" is not a valid email address

`email` duplicate within file	Row {N}: email "{value}" appears more than once in this file (also in row {M})
`email` duplicate in tenant	Row {N}: email "{value}" is already registered in your organisation
`phone` invalid format (if provided)	Row {N}: phone "{value}" contains invalid characters
`phone` duplicate within file	Row {N}: phone "{value}" appears more than once in this file (also in row {M})
`phone` duplicate in tenant	Row {N}: phone "{value}" is already registered in your organisation
`password` too weak (if provided)	Row {N}: password does not meet minimum requirements (8 chars, 1 uppercase, 1 number)
`department_name` not found	Row {N}: department "{value}" does not exist or is archived
`team_name` not found in dept	Row {N}: team "{value}" does not exist in department "{dept_value}"
`team_name` provided without `department_name`	Row {N}: team_name requires department_name to be set
`employee_roll_number` duplicate in tenant	Row {N}: roll number "{value}" is already assigned to another learner

6.5 Auto-Generated Passwords (CSV)

When `password` column is blank for a row:

- System generates a random 12-character password: uppercase + lowercase + digits + 1 special char
- Password is included in the **downloadable result CSV** in a `generated_password` column
- Result CSV is available for download for **48 hours** only (then purged)
- Client Admin is responsible for distributing passwords to learners
- Result CSV is **never emailed** — only downloadable from the platform (security constraint)

6.6 Error Log Format

Downloaded file: `keyskillset_import_errors_{timestamp}.csv`

Column	Content
`row_number`	Row number in original file
`full_name`	Value from file
`email`	Value from file
`phone`	Value from file
`error_message`	Human-readable error description

6.7 Acceptance Criteria

AC-2.2.1 — Valid CSV imports successfully (sync)

```
Given I am logged in as Client Admin
And I have a valid CSV with 50 rows, all unique emails and phones
When I upload the CSV
Then the system validates all 50 rows with 0 errors
And I see the validation result screen:
    Total rows: 50  |  Valid: 50  |  Errors: 0
And I click "Import Valid Rows Only (50 rows)"
Then all 50 learners are created synchronously
And I see success message: "50 learners added successfully."
And 50 audit entries are created: action = LEARNER_CREATED (bulk)
```

AC-2.2.2 — Mixed CSV: valid rows imported, error rows skipped

```
Given a CSV with 80 rows where:
    - Row 12: email already exists in tenant
    - Row 35: phone already exists in tenant
    - Row 67: department_name "XYZ Batch" does not exist
    - All other 77 rows are valid
When I upload the CSV and validation completes
Then validation result screen shows:
    Total rows: 80  |  Valid: 77  |  Errors: 3
And the error detail shows:
    Row 12: email "{value}" is already registered in your organisation
    Row 35: phone "{value}" is already registered in your organisation
    Row 67: department "XYZ Batch" does not exist or is archived
And "Download Error Log" button is visible
When I click "Import Valid Rows Only (77 rows)"
Then 77 learners are created
And rows 12, 35, 67 are not created
And I can download the error log with those 3 rows
```

AC-2.2.3 — Async import triggered for >100 rows

```
Given a valid CSV with 250 rows
When I upload the CSV and click "Import Valid Rows Only (250 rows)"
Then the system queues an async job
And I see: "Your import is processing. We'll email you at [admin email] when complete."
```

```
And I receive an email on job completion with import summary
And the result CSV (with any generated passwords) is downloadable from the platform for 48 hours
```

AC-2.2.4 — File size limit enforced

```
Given a CSV file of 6 MB
When I attempt to upload it
Then the system rejects it immediately:
    "File size exceeds the 5 MB limit. Please split your file and upload in batches."
And no rows are processed
```

AC-2.2.5 — Duplicate within file detected

```
Given a CSV where row 5 and row 23 both have email "duplicate@example.com"
When validation runs
Then both rows are flagged as errors:
    Row 5: email "duplicate@example.com" appears more than once in this file (also in row 23)
    Row 23: email "duplicate@example.com" appears more than once in this file (also in row 5)
And neither row is imported even if the email doesn't exist in the tenant
```

AC-2.2.6 — Auto-generated passwords available in result file

```
Given a CSV where 20 rows have blank password column
When the import completes successfully
Then a result CSV is generated with a "generated_password" column
And the result CSV is available for download for 48 hours
And the result CSV is NOT emailed
And Client Admin sees a banner:
    "Download the result file to retrieve auto-generated passwords.
     Available for 48 hours."
```

AC-2.2.7 — Team Manager CSV upload scoped to own dept

```
Given I am Team Manager for "NEET Batch 2026"
And my CSV has a row with department_name = "JEE Batch 2026"
When validation runs
Then that row is flagged:
    Row {N}: You can only add learners to your own department (NEET Batch 2026)
And the row is not imported
```

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7. Learner List View — Full Feature Spec

7.1 Client Admin Learner List

Entry: Learners (sidebar)

Columns:

Column	Sortable	Notes
Full Name	Yes	Links to learner profile
Email	No	

Phone	No	
Department	Yes	"Unassigned" if no dept
Team	No	"—" if no team
Status	Yes	Active / Inactive badge
Date Added	Yes	UTC date
Actions	No	View Profile / Deactivate or Reactivate / Reset Password

Filters (sidebar or top filter bar):

- Department (multi-select dropdown — active depts + "Unassigned")
- Status (Active / Inactive / All)
- Date Added (date range picker)

Search:

- Full-text search across: Full Name, Email, Phone, Employee/Roll Number
- Minimum 2 characters to trigger search
- Debounced 300ms

Pagination:

- Default 25 rows per page
- Options: 25 / 50 / 100
- Total count shown: "Showing 1–25 of 342 learners"

Bulk actions (checkbox select):

- Add to Group
- Deactivate selected (Client Admin only)
- Export selected as CSV

7.2 Team Manager Learner List

Identical to Client Admin list but:

- Filtered to own Department only at middleware
- No cross-dept filter shown
- No Deactivate action visible

- Bulk actions: Add to Group (V2 — TM cannot manage groups in V1), Export as CSV only

7.3 Acceptance Criteria

AC-2.3.1 — Search works across all searchable fields

```
Given I am Client Admin with 200 learners in my tenant
When I search "9876"
Then results show all learners where:
    full_name, email, phone, or employee_roll_number contains "9876"
And results update within 300ms of typing stopping
```

AC-2.3.2 — Department filter applies correctly

```
Given I filter by Department = "NEET Batch 2026"
Then only learners with department_id = NEET Batch 2026 are shown
And the count reflects the filtered set
```

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8. Learner Profile — Full Feature Spec

8.1 Profile Page Layout

Accessible via: Learner List → learner row → "View Profile" OR clicking the learner's name

Sections:

Section	Content
Header	Full Name, Status badge (Active/Inactive), Date Added, Created By
Basic Info	Email, Phone, Employee/Roll Number, Notes
Org Assignment	Department, Team, Groups (read-only list)
Learning Summary	Paths enrolled, paths completed, last active date (links to Reports)
Actions bar	Edit Profile / Reset Password / Deactivate or Reactivate

8.2 Edit Profile

Editable fields: Full Name, Phone, Employee/Roll Number, Notes, Department (Client Admin only), Team

Non-editable fields: Email (permanent identifier — V2 to support email change), Status (changed via dedicated Deactivate/Reactivate action, not via edit form)

Validation on edit:

- Same rules as creation
- Uniqueness checks exclude the current learner (editing own phone to same value is valid)

8.3 Acceptance Criteria

AC-2.4.1 — Edit profile updates correctly

```
Given learner Priya has phone "9876543210"
When I edit her profile and change phone to "9999999999"
And "9999999999" is not used by any other learner in the tenant
Then Priya's phone is updated to "9999999999"
And audit entry: action = LEARNER_UPDATED
    before_state = { phone: "9876543210" }
    after_state  = { phone: "9999999999" }
```

AC-2.4.2 — Email is non-editable

```
Given I am on Priya's edit profile form
Then the email field is read-only
And no API call to update email is possible via this form
```

AC-2.4.3 — Client Admin can change department from profile

```
Given Priya is in Department "NEET Batch 2026"
When I open Priya's profile → Edit Profile
And I change Department to "Foundation Batch"
Then Priya's department_id = Foundation Batch
And this is equivalent to the LEARNER_MOVED action from PRD 1
And audit entry: action = LEARNER_MOVED
```

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9. Learner Lifecycle Actions — Full Feature Spec

9.1 Deactivate Learner

Who can: Client Admin only.

Entry: Learner profile → "Deactivate" button OR Learner list → Actions → "Deactivate"

What deactivation does:

- Sets `user.status = INACTIVE`
- Learner **cannot log in** (authentication middleware blocks INACTIVE users)
- All active learning path enrollments are **paused** (progress retained, not deleted)
- Learner disappears from Team Manager's learner list
- Learner remains visible in Client Admin's list (filterable by Status = Inactive)
- Content assignments are **not deleted** (reactivation restores access)

What deactivation does NOT do:

- Does not delete the learner record
- Does not remove org assignments (dept, team, group memberships retained)
- Does not cancel certificates already earned
- Does not remove from audit log history

Confirmation modal:
"Deactivate Priya Sharma? She will lose access to the platform immediately. Her learning progress will be retained and can be restored if reactivated."

Acceptance Criteria:

AC-2.5.1 — Deactivation blocks login

```
Given learner Priya has status = ACTIVE
When I click Deactivate on Priya's profile and confirm
Then user.status = INACTIVE
And Priya's login attempt returns:
    "Your account has been deactivated. Contact your administrator."
And Priya's path enrollments retain their current progress (not deleted)
And Priya is no longer visible in Team Manager's learner list
And audit entry: action = LEARNER_DEACTIVATED
    actor = Client Admin
    entity = learner: Priya Sharma
```

AC-2.5.2 — Team Manager cannot deactivate

```
Given I am Team Manager for "NEET Batch 2026"
When I attempt: PATCH /api/learners/{priya_id}/deactivate
Then middleware returns 403 Forbidden
And Priya's status is unchanged
```

9.2 Reactivate Learner

Who can: Client Admin only.
Entry: Learner profile (of an INACTIVE learner) → "Reactivate" button

What reactivation does:

- Sets `user.status = ACTIVE`
- Learner can log in immediately
- All previously paused learning path enrollments become **active** again (progress exactly where left off)
- Learner reappears in Team Manager's learner list
- All org assignments (dept, team, group) are exactly as they were at deactivation

Acceptance Criteria:

AC-2.6.1 — Reactivation restores full access

```
Given learner Priya has status = INACTIVE
And Priya was at Step 2 of 4 in Learning Path "NEET Full Prep" before deactivation
When I click Reactivate on Priya's profile and confirm
Then user.status = ACTIVE
And Priya can log in immediately
And Priya's path enrollment shows Step 2 of 4 (progress intact)
And Priya reappears in her Team Manager's learner list
And audit entry: action = LEARNER_REACTIVATED
```

9.3 Password Reset by Client Admin

Who can: Client Admin (any tenant learner). Team Manager cannot reset passwords.

Entry: Learner profile → "Reset Password" button

Flow:

1. Client Admin clicks "Reset Password"
2. Modal appears with two fields: New Password + Confirm Password
3. Same password rules apply: min 8 chars, 1 uppercase, 1 number
4. On confirm: password hash updated immediately
5. No email sent to learner (admin-managed credential model)
6. Toast: *"Password reset for Priya Sharma."*
7. Audit log: `LEARNER_PASSWORD_RESET`

Acceptance Criteria:

AC-2.7.1 — Password reset works correctly

```
Given learner Priya exists with status = ACTIVE
When I click Reset Password on Priya's profile
And I enter new_password = "NewPass@456"
And I click Confirm
Then Priya's password hash is updated
And Priya can log in with "NewPass@456"
And Priya's old password no longer works
And no email is sent to Priya
And audit entry: action = LEARNER_PASSWORD_RESET
    actor = Client Admin
    entity = learner: Priya Sharma
    note: "password_hash updated; no email sent"
```

AC-2.7.2 — Team Manager cannot reset passwords

```
Given I am Team Manager for "NEET Batch 2026"
When I attempt: POST /api/learners/{priya_id}/reset-password
Then middleware returns 403 Forbidden
And Priya's password is unchanged
```

9.4 Remove Learner from Team

Who can: Client Admin (any team). Team Manager (own dept teams only).

Entry: Learner profile → Org Assignment section → "Remove from Team" button

OR: Department detail → Team detail → Learner row → "Remove from Team"

What it does:

- Sets `LearnerProfile.team_id = NULL`
- Learner remains in the parent Department
- Learning path enrollments and content assignments are unaffected
- Audit log: `LEARNER_REMOVED_FROM_TEAM`

Acceptance Criteria:

AC-2.8.1 — Remove from team retains department

```
Given Priya is in Department "NEET Batch 2026", Team "Batch A"
When I remove Priya from Team "Batch A"
Then Priya's team_id = NULL
And Priya's department_id = "NEET Batch 2026" (unchanged)
And Priya still appears in Department "NEET Batch 2026" learner list
And Priya's path enrollments are unaffected
And audit entry: action = LEARNER_REMOVED_FROM_TEAM
    before_state = { team_id: Batch A }
    after_state  = { team_id: NULL }
```

9.5 Remove Learner from Group

Who can: Client Admin only.

Entry: Learner profile → Org Assignment → Groups section → "Remove" next to group name

What it does:

- Soft-deletes `LearnerGroupMembership` record (`removed_at = NOW()`)
- Dept/Team assignment unaffected
- Future Group-based content assignments will not apply to this learner
- Existing content entitlements already created via Group assignment are **not revoked**

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10. Learner State Machine

```
LEARNER STATUS STATE MACHINE

  [Created] ──────────────────────────────► ACTIVE
      │                                        │
      │                                   deactivate (CA only)
      │                                        │
      │                                        ▼
      │                                    INACTIVE
```

```
      │                                        │
      │                                  reactivate (CA only)
      │                                        │
      └────────────────────────────────────────┘
                                          (cycle repeats)

  Note: Hard delete → V2
  Note: INACTIVE learner retains all data — enrollments, progress, certs
  Note: INACTIVE learner is invisible to Team Manager
  Note: INACTIVE learner is visible to Client Admin (status filter = Inactive)
```

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11. Complete Audit Event Reference

Action Constant	Trigger	Actor
`LEARNER_CREATED`	Single manual add	Client Admin or Team Manager
`LEARNER_CREATED_BULK`	CSV import (one entry per learner created)	Client Admin or Team Manager
`LEARNER_UPDATED`	Profile field changed	Client Admin or Team Manager
`LEARNER_DEACTIVATED`	Status set to INACTIVE	Client Admin
`LEARNER_REACTIVATED`	Status set to ACTIVE	Client Admin
`LEARNER_PASSWORD_RESET`	Admin resets password	Client Admin
`LEARNER_REMOVED_FROM_TEAM`	team_id set to NULL	Client Admin or Team Manager
`LEARNER_REMOVED_FROM_GROUP`	GroupMembership soft-deleted	Client Admin
`CSV_IMPORT_STARTED`	CSV job queued	Client Admin or Team Manager
`CSV_IMPORT_COMPLETED`	Async job finishes	System
`CSV_IMPORT_FAILED`	Async job fails (infrastructure error)	System

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12. Edge Cases & System Behaviour

Edge Case	System Behaviour
Add learner to a Department that is ARCHIVED	Blocked — "This department is archived. Select an active department."

CSV row has `department_name` that gets archived between upload and commit	Row validation re-checks at commit time; row fails if dept is now archived
Deactivate a learner who is mid-path	Status = INACTIVE; path enrollment paused; progress retained
Reactivate a learner whose Department was archived while inactive	Learner is reactivated; department_id = NULL (Unassigned); Client Admin sees Unassigned indicator
CSV file has all rows invalid	Validation result screen shows 0 valid rows; "Import Valid Rows Only" button is disabled
Client Admin uploads correct file type but wrong internal format (e.g. .csv extension but tab-separated)	Parser attempts auto-detect; if headers not found, returns error: "Could not parse file. Please download the template and use CSV format."
Team Manager uploads CSV with a learner already in another dept	Row fails: "This learner is already assigned to another department. Only Client Admin can reassign learners across departments." — Note: this only fires if the email/phone match an existing learner in a different dept
Two admins simultaneously add learner with same email	DB unique constraint catches the race condition; second insert returns 409 Conflict; UI shows: "This email was just registered by another admin. Please refresh and check the learner list."
Phone number entered with spaces or dashes (e.g. "98765 43210")	System normalises by stripping non-numeric chars (except leading `+`) before validation and storage
CSV result file (with generated passwords) not downloaded within 48 hours	File is purged; Client Admin must use password reset to set new passwords for those learners

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13. API Endpoints Reference

Learner Management

Method	Endpoint	Who	Description
`GET`	`/api/learners`	Client Admin, Team Manager (scoped)	List learners with filter/search/pagination
`POST`	`/api/learners`	Client Admin, Team Manager (own dept)	Create single learner
`GET`	`/api/learners/:id`	Client Admin, Team Manager (own dept)	Get learner profile
`PATCH`	`/api/learners/:id`	Client Admin, Team Manager (own dept)	Update learner profile
`POST`	`/api/learners/:id/deactivate`	Client Admin	Deactivate learner
`POST`	`/api/learners/:id/reactivate`	Client Admin	Reactivate learner
`POST`	`/api/learners/:id/reset-password`	Client Admin	Reset learner password
`DELETE`	`/api/learners/:id/team`	Client Admin, Team Manager (own dept)	Remove learner from team
`DELETE`	`/api/learners/:id/groups/:groupId`	Client Admin	Remove learner from group

Duplicate Check (real-time, called on blur)

Method	Endpoint	Who	Description
`GET`	`/api/learners/check-email`	Client Admin, Team Manager	Check email uniqueness within tenant
`GET`	`/api/learners/check-phone`	Client Admin, Team Manager	Check phone uniqueness within tenant

CSV Import

Method	Endpoint	Who	Description
`GET`	`/api/learners/csv-template`	Client Admin, Team Manager	Download CSV template
`POST`	`/api/learners/csv-validate`	Client Admin, Team Manager	Validate CSV, return per-row result
`POST`	`/api/learners/csv-import`	Client Admin, Team Manager	Commit import (sync or async)
`GET`	`/api/learners/csv-import/:jobId`	Client Admin, Team Manager	Poll async import status
`GET`	`/api/learners/csv-result/:jobId`	Client Admin, Team Manager	Download result CSV (48hr window)

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14. Success Metrics

Metric	Target
Time to manually add a single learner (form fill to save)	< 90 seconds
CSV validation response time for ≤100 rows	< 3 seconds
CSV import success rate (valid rows committed / valid rows detected)	100% — no partial commit failures on valid rows
Error log download availability after failed import	100% — always downloadable if errors exist
Zero cross-tenant learner data leaks	0 incidents
Team Manager cross-dept add attempts blocked (403, not 200)	100%
Deactivation blocks login within 1 request cycle	100% — no session carry-over post-deactivation

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15. Self-Critique — PRD 2 Specific

Weakness	Impact	Mitigation
No email sent to learner on account creation — Client Admin distributes credentials manually	Medium — error-prone at scale; credential loss risk	V2: invite email with set-password link. Document as known V1 limitation.
Email is non-editable — learner who changes email address is stuck	Medium — common in coaching centres (students use parent emails)	V2: email change flow with verification. Document limitation.
Async CSV result file (with generated passwords) purged after 48 hours	Low-Medium — if Client Admin misses it, must reset passwords individually	Show persistent banner until downloaded; send reminder email at 24hr mark.
Phone normalisation (stripping spaces/dashes) could cause silent mismatch if admin enters "98765 43210" and later searches "9876543210"	Low	Normalise at storage + normalise at search — consistent behaviour.
No learner self-service password reset in V1 — full dependency on Client Admin	Medium for large orgs	V2 priority. Client Admin password reset is the workaround.
Race condition on simultaneous add (two admins) handled at DB level (409) — UI error message may confuse	Low	Clear error message guides admin to refresh and check list.

End of PRD 2 — keySkillset Client Admin Platform: Learner Management v1.0